COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 10 : SLO – 2 | NAND gate and universal implementation
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Part B – Multiple Choice Questions

- 1. B — (A·B)'
- 2. B — It can implement AND, OR, NOT
- 3. B — Connect both inputs together
- 4. B — (A'·B')'
- 5. D — All of the above

Part A – Match the Following

- 1. NAND gate — c) Output is 0 only when all inputs are 1
- 2. NAND as NOT — a) Connect both inputs together
- 3. NAND as AND — e) Invert NAND output → (A·B)'' = A·B
- 4. NAND as OR — b) De Morgan: A + B = (A'·B')'
- 5. Universal Gate — d) Any logic function can be built using it

Part C – Activity Questions

- 1. 2-input NAND truth table: 00→1, 01→1, 10→1, 11→0.
- 2. NAND as NOT: tie both inputs together → Y = (A·A)' = A■. Truth table: 0→1, 1→0.
- 3. OR using only NAND gates: invert A → A', invert B → B' (two NANDs as NOTs), then NAND the results: (A'·B')' = A + B. Total 3 NAND gates.